LAB5

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1. Merge Sort

C++ Code

```
#include <iostream>
using namespace std;

void merge(int arr[], int low, int mid, int high)
{
    int i = low, j = mid + 1, k = 0;
    int temp[100];

    while (i <= mid && j <= high)
    {
        if (arr[i] <= arr[j])
            temp[k++] = arr[i++];
        else
            temp[k++] = arr[j++];
    }

    while (i <= mid)
        temp[k++] = arr[i++];

    while (j <= high)
        temp[k++] = arr[j++];

    for (i = low, k = 0; i <= high; i++, k++)
        arr[i] = temp[k];
}

void mergeSort(int arr[], int low, int high)
{
    if (low < high)
    {
        int mid = (low + high) / 2;
        mergeSort(arr, low, mid);
        mergeSort(arr, mid + 1, high);
        merge(arr, low, mid, high);
    }
}

int main()
{
    int n, arr[100];

    cout << "Enter number of elements: ";
    cin >> n;

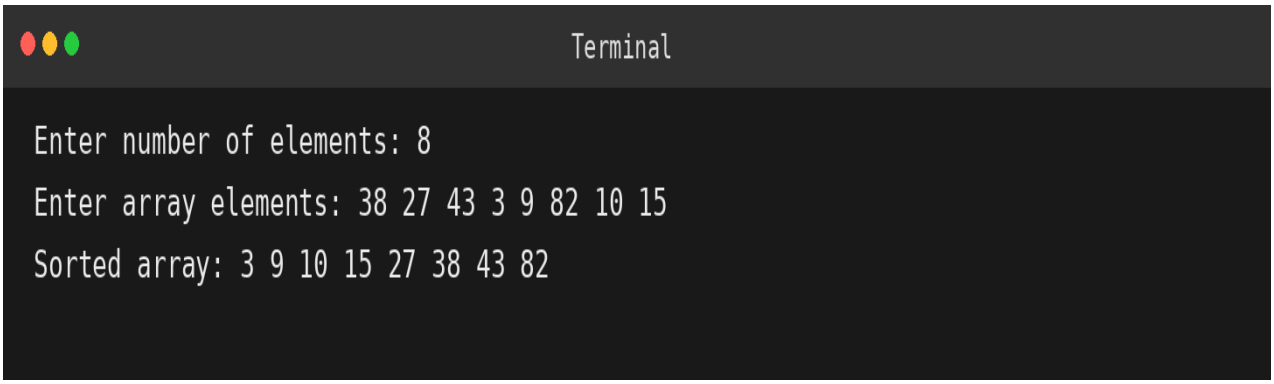
    cout << "Enter array elements: ";
    for (int i = 0; i < n; i++)
        cin >> arr[i];

    mergeSort(arr, 0, n - 1);

    cout << "Sorted array: ";
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";

    return 0;
}
```

Output


Terminal
Enter number of elements: 8
Enter array elements: 38 27 43 3 9 82 10 15
Sorted array: 3 9 10 15 27 38 43 82

2. Bucket Sort

C++ Code

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

void bucketSort(int arr[], int n)
{
    int maxValue = arr[0];

    for (int i = 1; i < n; i++)
        if (arr[i] > maxValue)
            maxValue = arr[i];

    vector<int> bucket[maxValue + 1];

    for (int i = 0; i < n; i++)
        bucket[arr[i]].push_back(arr[i]);

    int index = 0;

    for (int i = 0; i <= maxValue; i++)
    {
        sort(bucket[i].begin(), bucket[i].end());

        for (int j = 0; j < bucket[i].size(); j++)
            arr[index++] = bucket[i][j];
    }
}

int main()
{
    int n, arr[100];

    cout << "Enter number of elements: ";
    cin >> n;

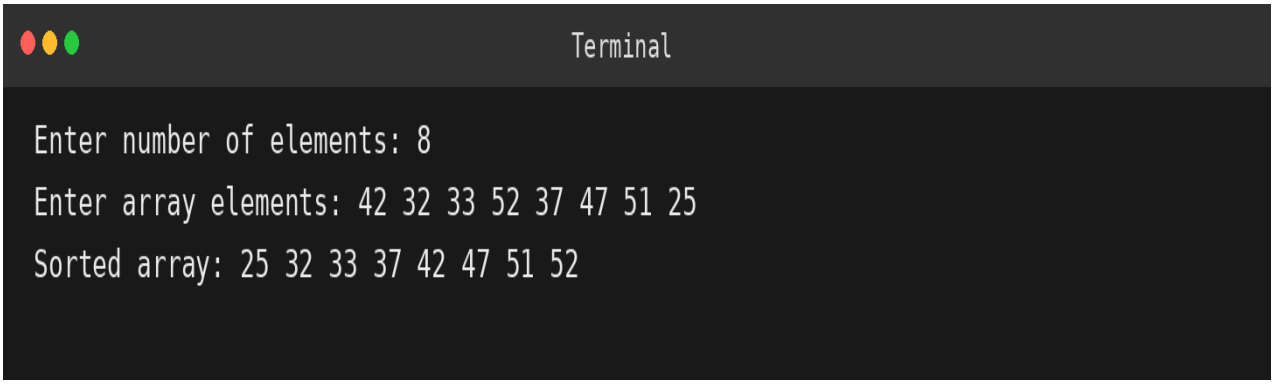
    cout << "Enter array elements: ";
    for (int i = 0; i < n; i++)
        cin >> arr[i];

    bucketSort(arr, n);

    cout << "Sorted array: ";
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";

    return 0;
}
```

Output



3. Quick Sort

C++ Code

```
#include <iostream>
using namespace std;

int partition(int arr[], int low, int high)
{
    int pivot = arr[high];
    int i = low - 1;

    for (int j = low; j < high; j++)
    {
        if (arr[j] < pivot)
        {
            i++;
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
        }
    }

    int temp = arr[i + 1];
    arr[i + 1] = arr[high];
    arr[high] = temp;

    return i + 1;
}

void quickSort(int arr[], int low, int high)
{
    if (low < high)
    {
        int p = partition(arr, low, high);
        quickSort(arr, low, p - 1);
        quickSort(arr, p + 1, high);
    }
}

int main()
{
    int n, arr[100];

    cout << "Enter number of elements: ";
    cin >> n;

    cout << "Enter array elements: ";
    for (int i = 0; i < n; i++)
        cin >> arr[i];

    quickSort(arr, 0, n - 1);

    cout << "Sorted array: ";
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";

    return 0;
}
```

Output

